

App Development Project Report

App Dev Project Report

1. Student Details

Name: Sandhya Gupta

Roll Number: 22f3000356

Email: 22f3000356@ds.study.iitm.ac.in

About Me: I am pursuing the IIT Madras BS Degree in Data Science and Applications. This project is a Placement Portal that automates the campus recruitment process through dedicated Admin, Company, and Student portals.

2. Project Details

Project Title: Placement Portal

Problem Statement:

To develop a web-based Placement Portal that digitizes campus recruitment by enabling administrators to manage companies and students, companies to conduct recruitment, and students to apply for placement drives.

Approach:

The application was developed using Flask for the backend and Vue.js for the frontend following a REST API architecture. JWT authentication provides secure role-based access. SQLite with SQLAlchemy manages relational data. Redis caching improves performance, while Celery and Celery Beat execute background jobs such as interview reminders, monthly reports, and asynchronous CSV exports.

3. AI/LLM Declaration

AI usage percentage: 40% usage of (help used in CSS Styling and VueJs implementation). Percentage calculated based on the metric provided "App Dev Project AI/LLM percentage check" document.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	ORM
Vue.js 3	Frontend
Bootstrap 5	Frontend styling and responsive design
Celery	Async Jobs
Celery Beat	Schedule Jobs
JWT	Authentication
SQLite	Lightweight local database for storing user data
Redis	Caching & Broker
Flask-Mail + Mailhog	Email

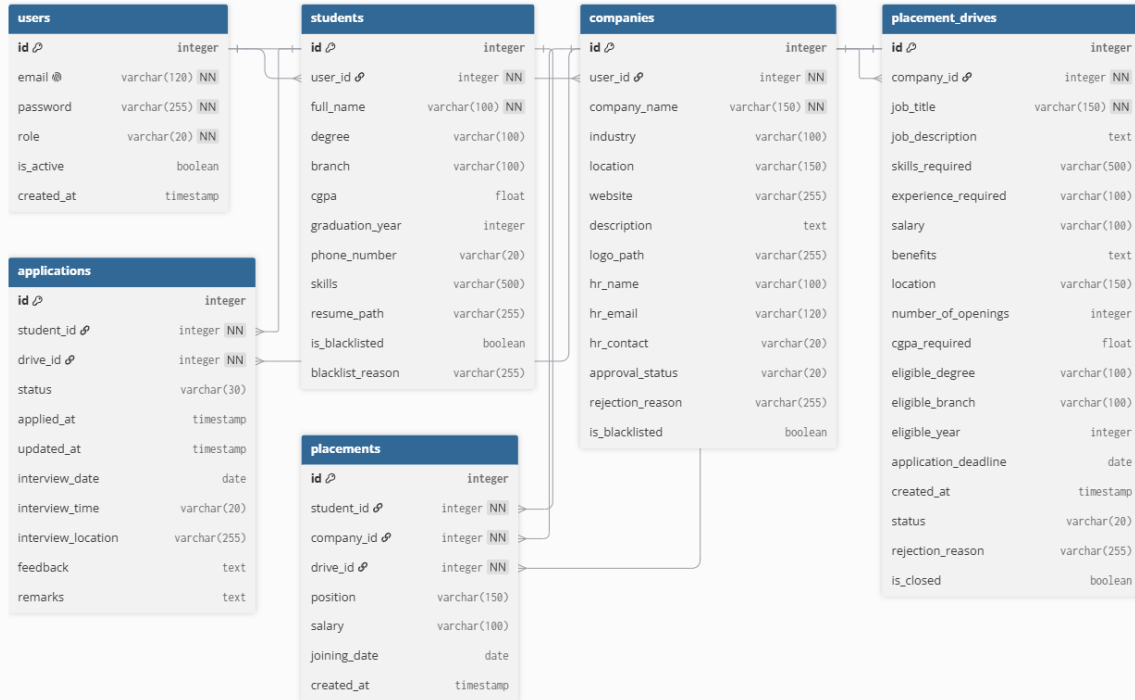
5. Database Schema / ER Diagram

Tables:

Users, Students, Companies, Placement Drives, Applications, Placements.

Relationships:

- User→Student
- User→Company
- Company→Placement Drives
- Student→Applications
- Placement Drive→Applications
- Placement records.



dbdiagram.io

6. API Resource Endpoints

Endpoint	Method	Description
/login	POST	Authenticate user and generate JWT token.
/student/register	POST	Register a new student account.
/company/register	POST	Register a new company account.
/admin/stats	GET	Retrieve admin dashboard statistics.
/admin/companies	GET	Retrieve all companies.
/admin/company/<company_id>	GET	Retrieve detailed company profile.
/admin/company/<company_id>/approve	PUT	Approve a company registration.

/admin/company/<company_id>/reject	PUT	Reject a company registration.
/admin/company/<company_id>/blacklist	PUT	Blacklist or deactivate a company.
/admin/company/<company_id>/restore	PUT	Restore a blacklisted company.
/admin/students	GET	Retrieve all students.
/admin/student/<student_id>	GET	Retrieve detailed student profile.
/admin/student/<student_id>/blacklist	PUT	Blacklist a student.
/admin/student/<student_id>/restore	PUT	Restore a blacklisted student.
/admin/drives	GET	Retrieve all placement drives.
/admin/drive/<drive_id>/approve	PUT	Approve a placement drive.
/admin/drive/<drive_id>/reject	PUT	Reject a placement drive.
/company/dashboard	GET	Retrieve company dashboard statistics.
/company/profile	GET	Retrieve company profile.
/company/drive	POST	Create a new placement drive.
/company/drives	GET	Retrieve all company placement drives.
/company/drive/<drive_id>	PUT	Update an existing placement drive.
/company/drive/<drive_id>/close	PUT	Close a placement drive.
/company/applications/<drive_id>	GET	Retrieve applicants for a placement drive.
/company/application/<application_id>/status	PUT	Update application status.
/company/student/<student_id>	GET	Retrieve a student's profile.
/company/export-applications	POST	Export company applications as CSV asynchronously.
/company/download-export/<filename>	GET	Download exported company CSV.
/student/dashboard	GET	Retrieve student dashboard information.
/student/profile	GET	Retrieve student profile.
/student/profile	PUT	Update student profile.

/student/resume	POST	Upload student resume.
/student/drives	GET	Retrieve approved placement drives.
/student/apply/<drive_id>	POST	Apply for a placement drive.
/student/applications	GET	Retrieve student applications and placement history.
/student/export-applications	POST	Export student applications as CSV asynchronously.
/student/download-export/<filename>	GET	Download exported student CSV.
/student/download-offer-letter/<application_id>	GET	Download placement offer letter as PDF.

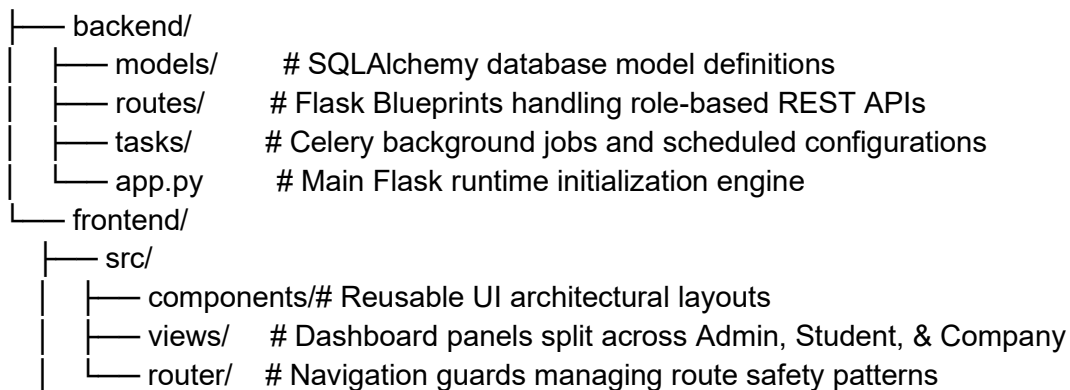
YAML API Definition

File:

Included separately in the submission ZIP as [api.yaml](#).

7. Architecture and Features

Architecture Overview:



7. Architecture and Features

- Role-based authentication using JWT for Admin, Company, and Student users.
 - Student and Company registration with secure login.
 - Admin approval and rejection of company registrations.
 - Company profile creation and management.
 - Student profile management with resume upload.
 - Creation, editing, approval, and closure of placement drives.
 - Search and filter placement drives based on job title, company, and degree.
 - Eligibility validation before allowing students to apply.
 - Prevention of duplicate job applications.
 - Application status tracking (Applied, Shortlisted, Interview Scheduled, Offer, Placed, and Rejected).
 - Interview scheduling with interview date, time, and location.
 - Company management of applicants including shortlisting, rejection, offer generation, and placement updates.
 - Student placement history with downloadable offer letter in PDF format.
 - Asynchronous CSV export of student applications and company applicants using Celery.
 - Automated interview reminder emails to students.
 - Monthly placement activity report generation for the administrator.
 - Redis-based caching to improve API performance.
 - Responsive web interface developed using Bootstrap 5.
-

8. Video Presentation

Drive Link:

<https://drive.google.com/file/d/1BTNj2pA27DmDCSgwaI5B0HDEmmWm9p2N/view?usp=sharing>
